

able or in the advanced stages of development. These included:

- Weed and pest control robots
- Seeding robots
- Fully autonomous tractors
- Fresh fruit and vegetable harvesting robots
- Agricultural drones
- Milking robots
- Autonomous carriers and platform robots

The most commercially developed robots are milking robots and agricultural drones. Milking robots can be operated indoors and under

unpredictable conditions. Cows can be easily trained to work with them.

Agricultural drones are best suited for imaging applications. Unmanned aerial vehicles (UAVs) can also be used for the delivery of organic herbicides to a field. The only issue here is that drones can only carry small payloads and have limited autonomy.

Another popular field robot is the weeding robot, which is meant for weed removal. The weeding robot helps eliminate manual weeding, which is quite labour-intensive.

Chemical usage in agriculture is not only harmful to the environment but also makes crops less organic. Precision agriculture is one of the best solutions to reduce this. It is a measurement strategy that processes and analyses temporal, spatial, and individual data and combines them to support management decisions, resource use efficiency, productivity, quality, profitability, and sustainability of agricultural production.

The robot also helps in cutting down the use of synthetic herbicides.

The state of development for a large diversity of depends upon how technically demanding it is. So, while milking robots are less complex to develop, making them highly commercially available, robots for fresh fruit and vegetable harvesting are undergoing research as they require complex AI systems for distinguishing crops at different stages of lifecycle and under different weather conditions.

Autonomous tractors: A possibility?

While it is nice to conceptualise the development of autonomous tractors thanks to all the talk going around advanced driver-assistance systems



Fig. 2: Milking robot (Credit: Flickr)

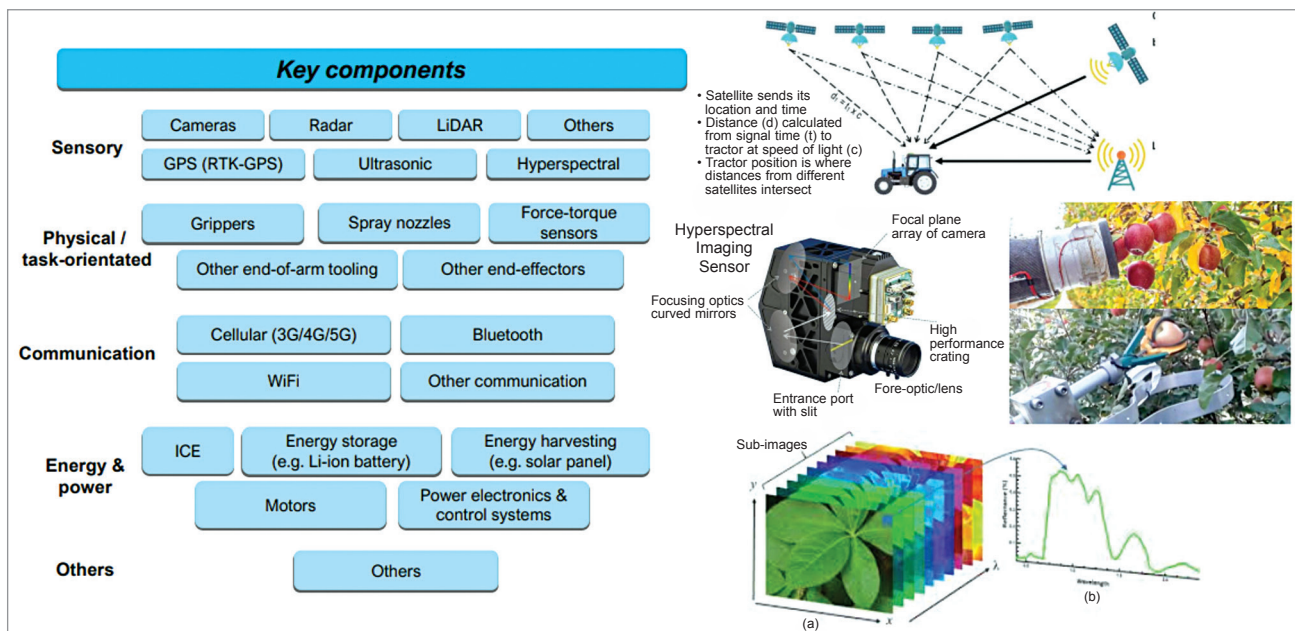


Fig. 3: Key components of agricultural robots (Credit: IDTechEx)